StegoVault: Algorithmic Architecture Manual

A Deep Dive into Cryptographic & Neural Network Algorithms

1. Cryptographic Algorithms

To guarantee complete confidentiality and tamper-proof security, StegoVault processes secret payloads through a robust cryptographic layer before performing steganographic embedding. The core algorithms include:

AES-GCM (Advanced Encryption Standard in Galois/Counter Mode): AES-GCM is an authenticated symmetric key algorithm. It encrypts the secret text using a 256-bit key. Crucially, it computes an authentication tag alongside the ciphertext. During decoding, this tag is verified. If a third party has modified the stego carrier (even by one byte), the validation checks fail, preventing silent payload corruption or tampering.

SHA-256 (Secure Hash Algorithm 256-bit): Used for key derivation and integrity mapping. The system hashes the user's password to create a secure key binding signature block: *Hash(Access Key + Carrier Signature)*. This binding is appended as metadata in the stego file footer, ensuring cryptographically secure identity verification.

2. Image Steganography: 2D Convolutional Neural Networks

The image modality utilizes an Encoder-Decoder CNN architecture modeled on the HiDDeN framework. The model contains three components:

A. Encoder Network: Takes the cover image tensor (3xHxW) and the message bit tensor (N) as input. It uses multiple convolutional layers with residual connections to project the message into the spatial feature map. The output is a stego image containing noise patterns that represent the hidden bits, but remain imperceptible to human vision (PSNR > 40 dB).

B. Decoder Network: A matching network that takes a stego image (or a distorted version of it) and applies spatial convolutions to reconstruct the original message bits. Spatial average pooling is applied at the output to yield a probability vector for each message bit.

C. Adversarial Discriminator: During training, a discriminator CNN acts as an adversary. It is trained to classify images as 'clean' or 'stego'. The encoder is optimized to deceive the discriminator, which forces the encoder to hide the message in highly complex textures, bypassing statistical steganalysis.

3. Audio Steganography: 1D CNNs & Psychoacoustic Masking

Audio carriers contain massive amounts of high-frequency data, but human ears are highly sensitive to phase shifts and random noise. To counter this, StegoVault employs two primary algorithms:

Phase-Hiding & Frequency Modulation: The 1D CNN processes audio signals in the frequency domain using Short-Time Fourier Transform (STFT). The algorithm embeds the secret payload by introducing microscopic phase shifts in the host audio frames, while leaving the magnitude spectrum unmodified. The changes are completely masked by dominant frequencies, preserving audio fidelity (SNR > 38 dB).

4. Video Steganography: 3D CNN Spatiotemporal Consistency

Video steganography requires managing both static images (frames) and motion. Using simple 2D image steganography frame-by-frame leads to temporal flickering (visual noise between frames).

3D Convolutional Neural Networks (3D CNN): The video steganography model utilizes 3D convolutions (sliding filters across frame sequences as a third dimension). This maintains temporal consistency, ensuring the stego noise changes smoothly with the optical flow of the video. The model places the stego payload in high-motion regions, where visual compression algorithms and human eyes are least likely to detect it.

5. Text Steganography: Semantic Transformer Encoding

Text files contain no pixels or audio waves. Embedding data in text requires natural language processing (NLP).

Transformer Attention & synonym sharding: The text steganography engine parses text documents using a pre-trained **Transformer Encoder** (with multi-head self-attention). The algorithm identifies tokens (words) with high semantic redundancy. It replaces these words with contextually matching synonyms derived from a shared dictionary (e.g. replacing 'begin' with 'start'). The replacement choices are governed by a Huffman coding tree, mapping message bits directly to synonym sequences without changing the meaning, grammar, or spelling of the document.

6. Architectural Summary of Algorithms

System Layer	Algorithm / Model Name	Mathematical Operator	Primary Objective
Cryptography	AES-256-GCM + SHA-256	Galois Field Auth Multiplication	Confidentiality & Integrity
Image Stego	ImageStegoNet (CNN)	2D Spatial Convolutions	Visual Imperceptibility
Audio Stego	AudioStegoNet (1D CNN)	STFT Frequency Modulations	Auditory Masking Fidelity
Video Stego	VideoStegoNet (3D CNN)	Spatiotemporal Convolutions	Temporal Flicker Prevention
Text Stego	TextStegoTransformer	Multi-Head Attention	Semantic Preservation